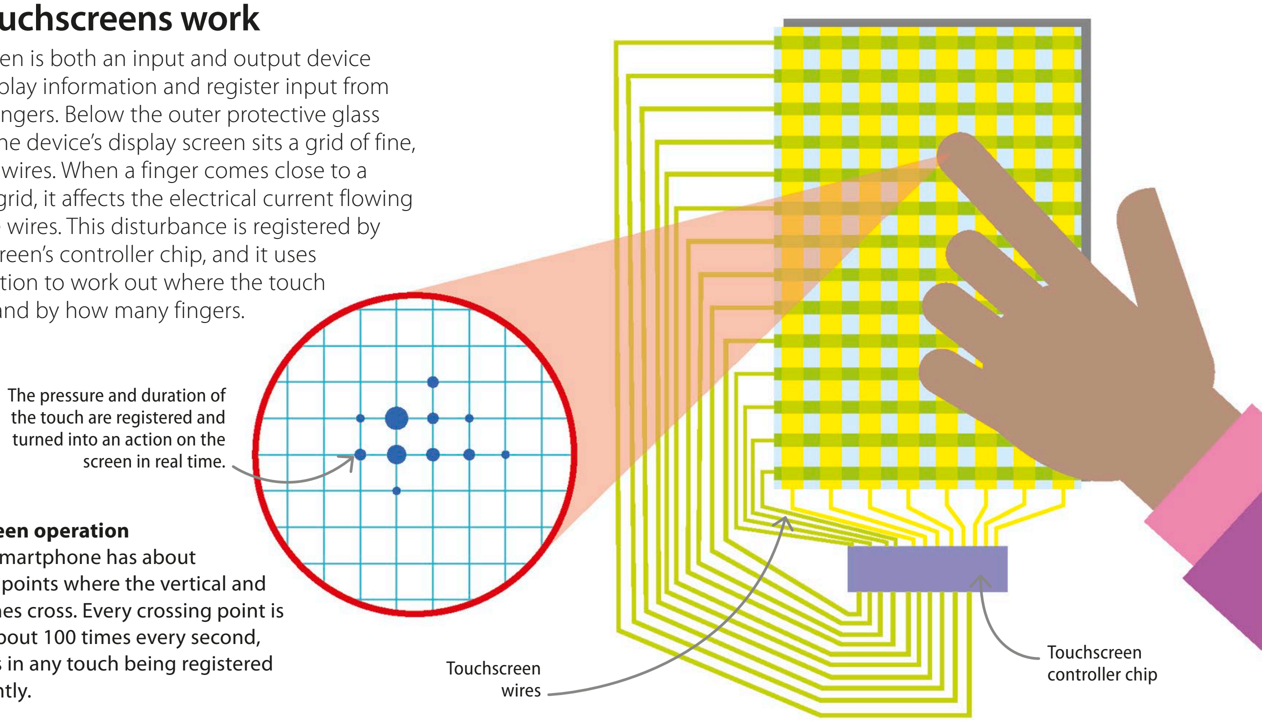
How touchscreens work

A touchscreen is both an input and output device that can display information and register input from a finger or fingers. Below the outer protective glass but above the device's display screen sits a grid of fine, transparent wires. When a finger comes close to a part of this grid, it affects the electrical current flowing through the wires. This disturbance is registered by the touchscreen's controller chip, and it uses this information to work out where the touch was made, and by how many fingers.

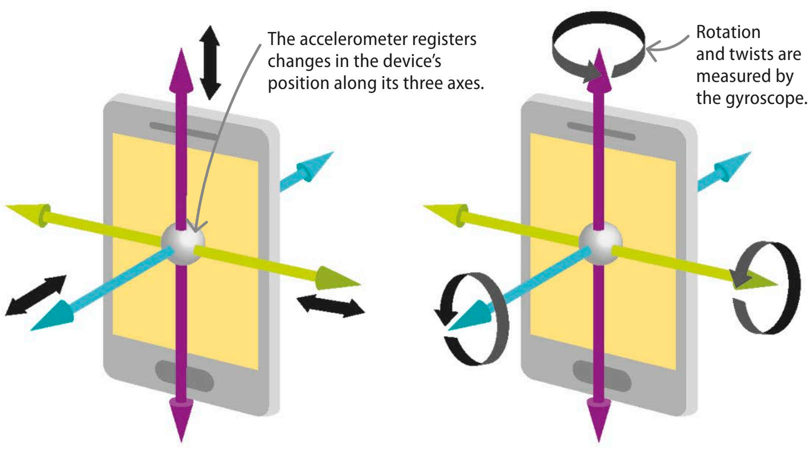


▷ **Touchscreen operation**
An average smartphone has about 150 crossing points where the vertical and horizontal lines cross. Every crossing point is monitored about 100 times every second, which results in any touch being registered almost instantly.

Tilt and twist

Smartphones and tablets detect changes in the orientation (position) of the phone. The accelerometer is a tiny chip that senses the tilting motion of the device. The gyroscope is a chip that adds more information to the accelerometer by measuring rotation or twists.

▽ **Accelerometer and gyroscope**
Accelerometers and gyroscopes are useful for changing the display of the device depending on how it is held, such as showing images in the right orientation, or as an additional input when playing games.



Accelerometer

Gyroscope

REAL WORLD

Voice navigation

Smartphones and tablets have become increasingly interactive and easier to use. A lot of the applications on a modern smartphone or tablet can understand voice commands, and react to them in real time. This is a huge help to disabled people who might not be able to use a touchscreen.